LAB ASSIGNMENT 15.2

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Task Descriptor-1: (Setup Fast API Backend)
Instructions:
• Install Fas tAPI and Uvicorn.
• Use AI to generate a basic FastAPI application with one endpoint
Sample Code:
from fastapi import FastAPI
app = FastAPI()
@app.get("/")
def read_root():
return {"message": "Welcome to AI-powered FastAPI"}
Expected Output:
Running the server and accessing:
http://127.0.0.1:8000/
Response:
{"message": "Welcome to AI-powered FastAPI"}

Code:

```
main.py > read_root
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def read_root():
    return {"message": "Welcome to AI-powered FastAPI"}
```

Output:

```
PS C:\Users\bayya\Desktop\fastapi_project> & c:\Users\bayya\Desktop\fastapi_project\venv\Scripts\Activate.ps1
(venv) PS C:\Users\bayya\Desktop\fastapi_project>  & 'c:\Users\bayya\Desktop\fastapi_project\venv\Scripts\python.exe' 'c:\Users\bayya\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '64030' '--' 'C:\Users\bayya\Desktop\fastapi_project\main.py'
(venv) PS C:\Users\bayya\Desktop\fastapi_project> uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\bayya\\Desktop\\fastapi_project']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [16744] using StatReload
INFO:     Started server process [23764]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

127.0.0.1:8000

Pretty print

{"message":"Welcome to AI-powered FastAPI"}

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET / Read Root

Parameters

Try it out

No parameters

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body

```
{
  "message": "Welcome to AI-powered FastAPI"
}
```

Download

Response headers

```
content-length: 43
content-type: application/json
date: Mon,16 Mar 2026 15:12:17 GMT
server: uvicorn
```

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body

```
{
  "message": "Welcome to AI-powered FastAPI"
}
```

Download

Response headers

```
content-length: 43
content-type: application/json
date: Mon,16 Mar 2026 15:12:17 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Task Descriptor- 2: (API – Create and Read Items)

Instructions:

• Implement endpoints to add and retrieve products.

• Use a Python list as temporary storage.

Sample Code:

```
items = []
@app.get("/items")
def get_items():
return items
@app.post("/items")
def add_item(item: dict):
items.append(item)
return {"message": "Item created", "item": item}
```

Expected Output:

GET /items → []

POST /items with {"name":"Pen","price":20}

→ {"message":"Item created","item":{"name":"Pen","price":20}}

Code:

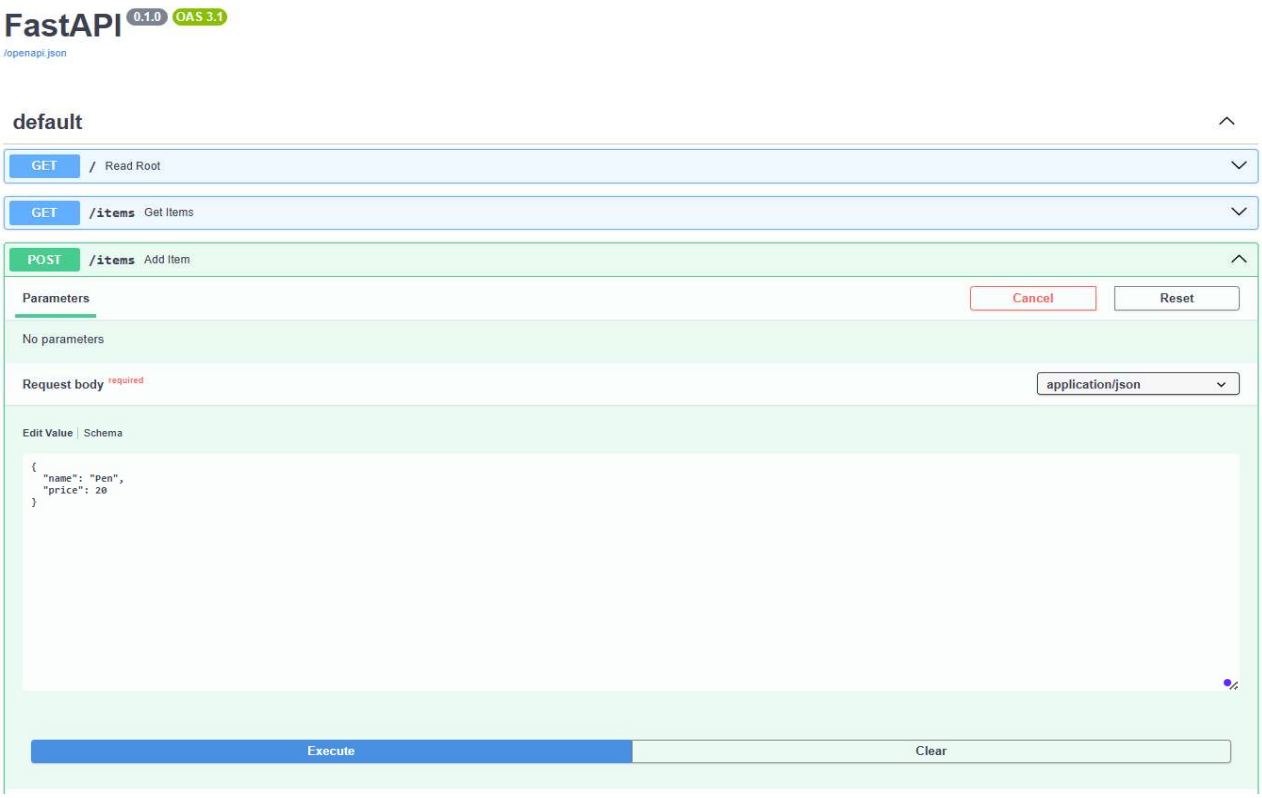
Welcome | main.py 1

main.py > get_items

```python
from fastapi import FastAPI
app = FastAPI()
# temporary storage
items = []
@app.get("/")
def read_root():
    return {"message": "Welcome to AI-powered FastAPI"}
@app.get("/items")
def get_items():
    return items
@app.post("/items")
def add_item(item: dict):
    items.append(item)
    return {"message": "Item created", "item": item}
```

Output:

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

(venv) PS C:\Users\bayya\Desktop\fastapi_project> uvicorn main:app --reload
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [5536]
INFO:     Started server process [4008]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:62285 - "GET / HTTP/1.1" 200 OK
INFO:     127.0.0.1:62285 - "GET / HTTP/1.1" 200 OK
INFO:     127.0.0.1:62447 - "GET /docs HTTP/1.1" 200 OK
INFO:     127.0.0.1:62447 - "GET /openapi.json HTTP/1.1" 200 OK
INFO:     127.0.0.1:60246 - "POST /items HTTP/1.1" 200 OK
INFO:     127.0.0.1:60246 - "POST /items HTTP/1.1" 200 OK
```



Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Pen",
  "price": 20
}'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code	Details
200	Response body

```
{
  "message": "Item created",
  "item": {
    "name": "Pen",
    "price": 20
  }
}
```

Download

Response headers

```
content-length: 59
content-type: application/json
date: Mon,16 Mar 2026 15:28:25 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Code	Description	Links
422	Validation Error	No links

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Task Descriptor- 3: (Update Existing Item)

Instructions:

• Use AI to generate an endpoint to update an item using its index.

Sample Code:

```
@app.put("/items/{index}")
def update_item(index: int, item: dict):
if index >= len(items):
return {"error": "Item not found"}
items[index] = item
return {"message": "Item updated", "item": item}
```

Expected Output:

PUT /items/0 with {"name":"Pencil","price":10}

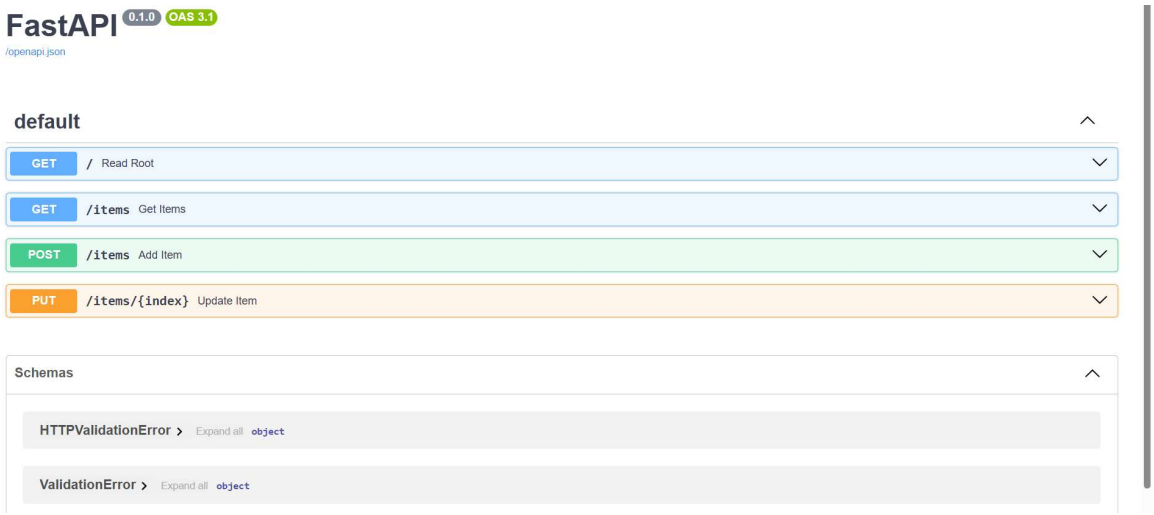
→ {"message":"Item updated","item":{"name":"Pencil","price":10}}

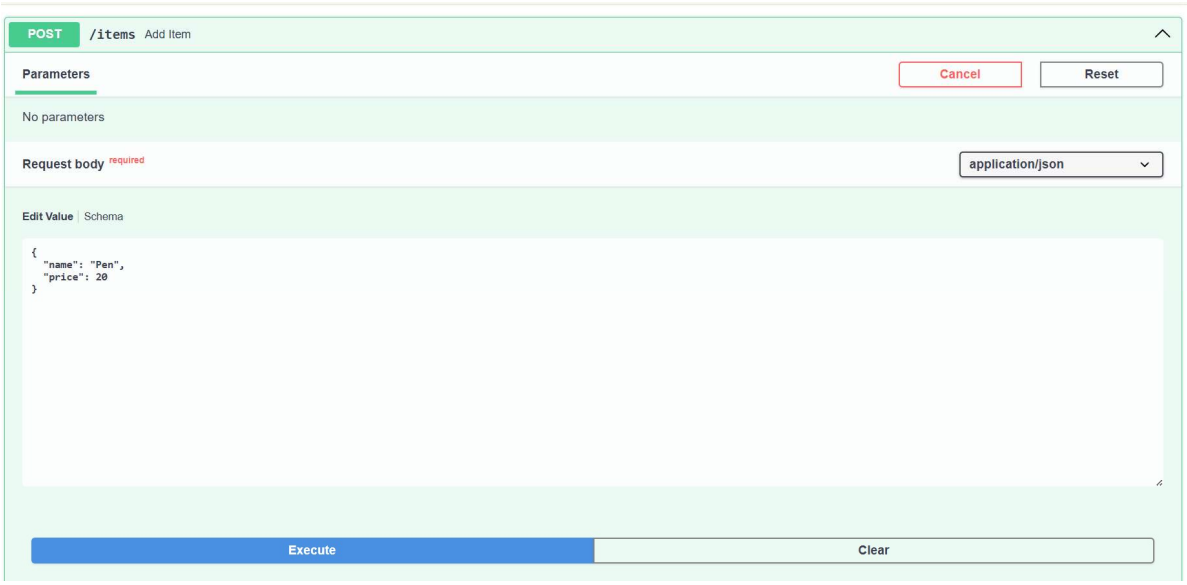
Code:

```python
from fastapi import FastAPI

app = FastAPI()

# temporary storage
items = []

# root endpoint
@app.get("/")
def read_root():
    return {"message": "Welcome to AI-powered FastAPI"}

# get all items
@app.get("/items")
def get_items():
    return items

# add item
@app.post("/items")
def add_item(item: dict):
    items.append(item)
    return {"message": "Item created", "item": item}

```

```python
# add item
@app.post("/items")
def add_item(item: dict):
    items.append(item)
    return {"message": "Item created", "item": item}

# update item using index
@app.put("/items/{index}")
def update_item(index: int, item: dict):
    if index >= len(items):
        return {"error": "Item not found"}

    items[index] = item
    return {"message": "Item updated", "item": item}
```

Output:


FastAPI
0.1.0
OAS 3.1
/openapi.json
default
GET / Read Root
GET /items Get Items
POST /items Add Item
PUT /items/{index} Update Item
Schemas
HTTPValidationError Expand all object
ValidationError Expand all object


POST /items Add Item
Parameters
Cancel
Reset
No parameters
Request body required
application/json
Edit Value | Schema
{
"name": "Pen",
"price": 20
}
Execute
Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Pen",
  "price": 20
}'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code	Details
200	

Response body

```
{
  "message": "Item created",
  "item": {
    "name": "Pen",
    "price": 20
  }
}
```

Download

Response headers

```
content-length: 59
content-type: application/json
date: Mon,16 Mar 2026 15:37:53 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

PUT /items/{index} Update Item

Parameters

Cancel Reset

Name	Description
index * required integer (path)	0

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/items/0' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Pencil",
  "price": 10
}'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

Code	Details
200	Response body

```
{
  "message": "Item updated",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Mon,16 Mar 2026 15:39:57 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header.	*No links*

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value | Schema "string"	No links

422	Validation Error
	Media type
	application/json
	Example Value | Schema
	{
	"detail": [
	{
	"loc": [
	"string",
	0
	],
	"msg": "string",
	"type": "string",
	"input": "string",
	"ctx": {}
	}
	]
	}
		No links

Task Descriptor- 4: (Delete Item)

Instructions:

• Implement deletion of an item by index.

Sample Code:

```
@app.delete("/items/{index}")
def delete_item(index: int):
if index >= len(items):
return {"error": "Item not found"}
removed = items.pop(index)
return {"message": "Item removed", "item": removed}
```

Expected Output:

DELETE /items/0

→ {"message":"Item removed","item":{"name":"Pencil","price":10}}

Code:

Welcome main.py 1

main.py > delete_item

```python
# update item
@app.put("/items/{index}")
def update_item(index: int, item: dict):
    if index >= len(items):
        return {"error": "Item not found"}

    items[index] = item
    return {"message": "Item updated", "item": item}

# delete item
@app.delete("/items/{index}")
def delete_item(index: int):
    if index >= len(items):
        return {"error": "Item not found"}

    removed = items.pop(index)
    return {"message": "Item removed", "item": removed}
```

Output:

Welcome main.py 1

main.py > delete_item

```python
# update item
@app.put("/items/{index}")
def update_item(index: int, item: dict):
    if index >= len(items):
        return {"error": "Item not found"}

    items[index] = item
    return {"message": "Item updated", "item": item}

# delete item
@app.delete("/items/{index}")
def delete_item(index: int):
    if index >= len(items):
        return {"error": "Item not found"}

    removed = items.pop(index)
    return {"message": "Item removed", "item": removed}
```

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET	/	Read Root
GET	/items	Get Items
POST	/items	Add Item
PUT	/items/{index}	Update Item
DELETE	/items/{index}	Delete Item

POST /items Add Item

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Pencil",
  "price": 10
}'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code	Details
200	Response body

```
{
  "message": "Item created",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Mon,16 Mar 2026 15:48:22 GMT
server: uvicorn
```

Responses

Code	Description	Links

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

DELETE `/items/{index}` Delete Item

Parameters Cancel

Name	Description
index * required `integer` *(path)*	0

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/items/0' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

Code	Details

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/items/0' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

Code	Details
200	Response body

```
{
  "message": "Item removed",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Mon,16 Mar 2026 15:49:04 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Task Descriptor -5: (Add Auto-Generated Documentation)

Instructions:

• Use AI to add inline comments and docstrings for all endpoints.

• Optionally, integrate Swagger / Flask-RESTX to auto-generate API documentation.

Sample Code:

```
@app.route('/items', methods=['GET'])
def get_items():
"""
GET /items
Returns a list of all items in the store.
```

Code:

```python
from fastapi import FastAPI
# Create FastAPI application
app = FastAPI(
    title="Product Management API",
    description="A simple FastAPI CRUD application to manage store items",
    version="1.0"
)
# Temporary storage for items
items = []
@app.get("/")
def read_root():
    """
    GET /

    Returns a welcome message for the API.
    """
    return {"message": "Welcome to AI-powered FastAPI"}
@app.get("/items")
def get_items():
    """
    GET /items

    Returns a list of all items stored in memory.
    """
    return items
@app.post("/items")
def add_item(item: dict):
    """
    POST /items

    Adds a new item to the store.

```

Welcome | main.py 1

main.py > get_items

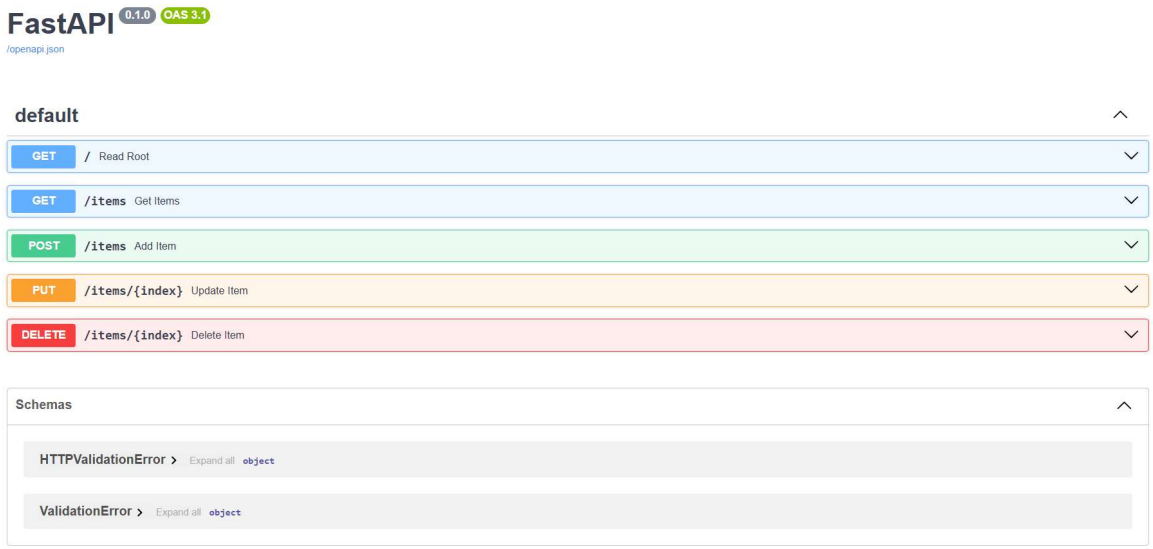
```python
@app.post("/items")
def add_item(item: dict):
    """
    POST /items

    Adds a new item to the store.

    Parameters:
    item (dict): JSON object containing item details such as name and price.

    Returns:
    JSON message confirming item creation.
    """
    items.append(item)
    return {"message": "Item created", "item": item}


@app.put("/items/{index}")
def update_item(index: int, item: dict):
    """
    PUT /items/{index}

    Updates an existing item using its index.

    Parameters:
    index (int): Position of the item in the list.
    item (dict): Updated item data.

    Returns:
    JSON message confirming item update.
    """
    if index >= len(items):
```

Welcome main.py 1

main.py > get_items

```python
def update_item(index: int, item: dict):
    if index >= len(items):
        return {"error": "Item not found"}

    items[index] = item
    return {"message": "Item updated", "item": item}


@app.delete("/items/{index}")
def delete_item(index: int):
    """
    DELETE /items/{index}

    Deletes an item from the list using its index.

    Parameters:
    index (int): Position of the item to remove.

    Returns:
    JSON message confirming deletion.
    """
    if index >= len(items):
        return {"error": "Item not found"}

    removed = items.pop(index)
    return {"message": "Item removed", "item": removed}


```

Output:


FastAPI
0.1.0
OAS 3.1
/openapi.json
default
GET / Read Root
GET /items Get Items
POST /items Add Item
PUT /items/{index} Update Item
DELETE /items/{index} Delete Item
Schemas
HTTPValidationError Expand all object
ValidationError Expand all object